

Chapter 5: Coordination Compounds

Reference Sheet + Tips & Tricks

Formula Sheet

I. IUPAC Naming – Rules and Order

Order of Naming a Coordination Compound

1. Name of the counter ion is written first, regardless of whether it appears before or after the parenthesis (coordination sphere) in the formula.
2. Ligand names are written in alphabetical order if more than one ligand is present.
3. Followed by the name of the central metal atom, then its oxidation number in roman numerals in brackets.
4. If the complex ion carries a negative or positive charge, the word “ion” is written after everything.
5. If the coordination sphere is anionic, the central metal atom’s name ends in “-ate”.

Cationic complex: No. and name of ligands (alphabetical) → Name of central metal atom → O.S. in roman → Name of anion.

Anionic complex: Name of cation → Number and name of ligands → Name of C.M.A. with “-ate” → O.S. in roman.

Examples:

$K_3[Fe(CN)_6]$ = potassium hexacyanidoferrate(III)

$[Fe(H_2O)_6]SO_4$ = hexaaquairon(II) sulphate

II. Ligand Reference Tables

Important Unidentate Ligands

H_2O – aqua NH_3 – ammine CO – carbonyl NO – nitrosyl C_5H_5N – pyridine F^- – fluoro
 Cl^- – chlorido Br^- – bromido NH_2^- – imido I^- – iodido CN^- – cyanido OH^- – hydroxo
 NO_2^- – nitro ONO^- – nitrito CH_3COO^- – acetato SCN^- – thiocyanato NCS^- – isothiocyanato

Important Didentate Ligands

$CH_2NH_2-CH_2NH_2$ – ethylenediamine (en) $C_2O_4^{2-}$ – oxalato $EDTA^{4-}$ – ethylenediaminetetraacetate
 $S_2O_3^{2-}$ – thiosulphato

III. Isomerism in Coordination Compounds

- **Ionisation isomerism** – exchange of ligand with counter ion: $[PtCl_2(NH_3)_4]Br_2 \leftrightarrow [PtBr_2(NH_3)_4]Cl_2$.
- **Hydrate isomerism** – the aqua (water) ligand exchanges with the outside counter ion:



- **Linkage isomerism** – the point of attachment of an ambidentate ligand changes: $[\text{Cr}(\text{SCN})(\text{H}_2\text{O})_5]^{2+} \leftrightarrow [\text{Cr}(\text{NCS})(\text{H}_2\text{O})_5]^{2+}$.
- **Coordination isomerism** – ligands are exchanged between the cationic and anionic complex ions of a compound: $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{C}_2\text{O}_4)_3] \leftrightarrow [\text{Co}(\text{C}_2\text{O}_4)_3][\text{Cr}(\text{NH}_3)_6]$.

IV. Valence Bond Theory – Hybridisation Reference

Coordination Number 4

sp^3 – Tetrahedral, e.g., $[\text{Ni}(\text{CO})_4]$, $[\text{MnCl}_4]^{2-}$, $[\text{ZnCl}_4]^{2-}$, $[\text{NiCl}_4]^{2-}$
 dsp^2 – Square Planar, e.g., $[\text{Pt}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{Cu}(\text{CN})_4]^{3-}$

Coordination Number 6

d^2sp^3 (inner orbital) – Octahedral, e.g., $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{PtCl}_6]^{2-}$
 sp^3d^2 (outer orbital) – Octahedral, e.g., $[\text{Cr}(\text{NH}_3)_6]^{3+}$

Steps to Find Hybridisation

1. Identify the coordination number of the compound.
2. Write the electronic configuration of the central metal atom.
3. Find the oxidation number of the central metal atom, then write the electronic configuration of its ion.
4. Identify weak or strong ligand, then correct the electron arrangement by strong (pairing) or weak (no pairing) attachment.
5. Fill the electrons donated by the ligands into the remaining vacant orbitals and read off the hybridisation.

Magnetic behaviour: vacant/unpaired orbitals $\rightarrow$ paramagnetic; all-paired/empty orbitals $\rightarrow$ diamagnetic.

Spin: weak ligand $\rightarrow$ high spin; strong ligand $\rightarrow$ low spin.

V. Crystal Field Theory

The d -orbitals of the coordination compound split into two sets: t_{2g} (lower energy) and e_g (higher energy).

How electrons are filled: (1) write the electronic configuration of the C.M.A.'s ion after removing its oxidation number's electrons; (2) identify ligand strength – a strong ligand causes pairing in t_{2g} before occupying e_g ; a weak ligand fills t_{2g} and e_g singly first (Hund's rule) before pairing.

Tips, Tricks & Hacks

- Counter-ion placement rule: if the counter ion is written *before* the parenthesis in the formula, the complex ion (in brackets) is negative and the counter ion is positive; if *after*, the complex ion is

positive and the counter ion is negative.

- Important electronic configurations to memorise: Sc(21) $3d^1 4s^2$, Ti(22) $3d^2 4s^2$, V(23) $3d^3 4s^2$, Cr(24) $3d^5 4s^1$, Mn(25) $3d^5 4s^2$, Fe(26) $3d^6 4s^2$, Co(27) $3d^7 4s^2$, Ni(28) $3d^8 4s^2$, Cu(29) $3d^{10} 4s^1$, Zn(30) $3d^{10} 4s^2$.
- Important ion configurations: $\text{Fe}^{2+} \rightarrow 3d^6$, $\text{Fe}^{3+} \rightarrow 3d^5$; $\text{Mn}^{2+} \rightarrow 3d^5$; $\text{Co}^{2+} \rightarrow 3d^7$, $\text{Co}^{3+} \rightarrow 3d^6$; $\text{Ni}^{2+} \rightarrow 3d^8$; $\text{Cu}^+ \rightarrow 3d^{10}$, $\text{Cu}^{2+} \rightarrow 3d^9$; $\text{Cr}^{3+} \rightarrow 3d^3$.